

Chua Yi Long

Lead Data Scientist / Lead AI Scientist — Agentic AI

yilongchua2705@gmail.com | [LinkedIn](#) | [GitHub](#) | [Medium](#) | Singaporean

Professional Summary

Yi Long Chua is building the next layer of reliable AI systems: agentic harnesses, model-evaluation workflows, and production ML platforms that turn prototypes into GTM products. Combines hands-on LLM training and evaluation, computer vision/OCR deployment, and cloud-native engineering with technical leadership across enterprise, government, and regulated financial-services environments.

Total Working Experience: (8 Years)

Accenture — AI Refinery for Advanced AI | Lead Data Scientist (Assistant Manager) | Dec 2024 – Present

Senior Data Scientist — Financial Institution

- Designed, built, and deployed a suite of production-ready AI agents supporting critical business functions in a regulated financial-services environment.
- Implemented a custom agent-to-agent communication protocol to improve cross-agent collaboration, task handoff, and workflow coordination.
- Contributed across agent architecture, orchestration logic, deployment readiness, and stakeholder-facing delivery.

Data Scientist — Internal AI Asset

- Designed and developed **MoCAG Unified Response Generation**, an internal enterprise AI asset for response generation workflows.
- Prototyped an architecture using a mixture-of-experts-based CAG model to improve enterprise response generation and reusable AI capability development.
- Worked across LLM workflow design, experimentation, and internal platformization for enterprise use cases.

Technical Lead Data Scientist — Government Agency, Generative AI

- Served as technical lead for a generative unit-test copilot for React.js applications.
- Designed the end-to-end architecture covering code parsing, prompt execution, generation workflow, and automated unit-test creation.
- Translated an ambiguous developer-productivity problem into a practical AI system architecture aligned with engineering workflows.

Patsnap — R&D Team | Data Scientist — Computer Vision & LLM | Feb 2022– Nov 2024

- Built and deployed two production model pipelines composed of multiple deep-learning models to detect properties, units, and measurements in images from patents and research papers, improving search and data-curation workflows.
- Fine-tuned Tesseract OCR 5.1 for domain-specific character, unit, and superscript recognition, achieving <5% character error rate (CER).
- Trained LLMs including Llama 2 and Mixtral-7B using **continual pretraining (CPT)** and **supervised fine-tuning (SFT)** for in-house data curation and evaluation tooling.
- Applied both **IA3 parameter-efficient fine-tuning** and full fine-tuning methods to compare cost, performance, and quality trade-offs across LLM development cycles.
- Implemented evaluation workflows for non-chat CPT/SFT models and chat SFT models across summarization, named-entity recognition, and zero-shot classification tasks.
- Improved model-training throughput through distributed-training optimization, balancing tensor parallelism and model parallelism while tuning chunking and batch-size strategy during CPT stages.
- Built and deployed multiple end-to-end computer-vision systems, including image classification, object detection, text detection, and OCR projects, from proof of concept to production on AWS infrastructure.

ENCO Technologies | Data Scientist — Computer Vision | Nov 2021– Nov 2023 (Remote – Contract)

- Designed and deployed automated visual quality-inspection systems for defect detection in industrial plastics manufacturing.
- Fine-tuned object-detection models for crack and scratch detection under varied production and environmental conditions.
- Integrated IoT systems with ML models and mobile interfaces to support initial field delivery and production-readiness.
- Partnered with engineering stakeholders to adapt CV models to real-world manufacturing constraints, including lighting variation, object variation, and deployment integration.

Tongdun International | Data Scientist — Credit Risk | Jan 2020– Nov 2021

- Served as project technical lead in a banking joint venture for credit-risk modelling, managing model development and deployment in accordance with regulatory requirements.
- Developed credit-risk models using alternative data to support broader financial inclusion for P2P lending platforms.
- Worked across model design, feature development, validation, stakeholder communication, and deployment readiness for regulated financial use cases.

Agentic AI Projects

CapyHome — Local AI Harness Build for deep research

- Design and build chat based Agentic harness targeted for long-running task (deep research)
- web-UI interface designed around Plan, Work mode with interactive panel (VS code style: Navigation panel with preview and editor)
- Build evaluation pipeline for testing prompts and evaluate latency and overall performance

Local Websearch Tool — SearxNG + crawl4ai – Integrated into CapyHome

- Built a localhost-first web-search and crawling toolchain with no dependency on paid search APIs.
- Integrated **SearxNG** for metasearch and **crawl4ai** for crawl/extraction workflows to support retrieval-enabled agents and local AI research workflows.
- Designed the stack as a MCP (tool call) with ability to increase parallel request through reverse load balancing for web search and crawling

LLM Wiki — Karpathy-Style Local Knowledge Build – Integrated into CapyHome

- Built a local LLM knowledge/wiki workflow inspired by Andrej Karpathy-style knowledge-building methods.
- Focused on creating a reliable local cached knowledge base for LLM to retained web search data for chat based
- Chrome Browser based Extensions for web clipping to bypass login credentials

Language Learning App — Dutch

- Build and developed 4 Distinct components [News Feed, Writing, Listening, Speaking]:
 - News Feed (Instagram style with news stories in dutch and English -translated)
 - Utilized XTTS v2 for voice generation and comfyUI for image generation on local systems
 - LLM based evaluation metrics in accordance with A2/B1 Dutch Exam format.
 - Deployed using zrok for external connections for mobile/on the go learning

Core AI / ML Strengths

- **Agentic AI & LLM Systems:** agent harness design, multi-agent collaboration, tool-using agents, prompt/workflow orchestration, LLM application architecture, internal AI assets.
- **LLM Training & Evaluation:** continual pretraining, supervised fine-tuning, PEFT/IA3, full fine-tuning, non-chat and chat model evaluation, summarization, NER, zero-shot classification, model behavior analysis.
- **Retrieval & Tooling:** local web search, crawling, retrieval-enabled workflows, knowledge-base construction, AI-assisted browsing, grounding workflows.
- **Computer Vision & OCR:** object detection, image classification, defect detection, patent/image OCR, Tesseract fine-tuning, crack/scratch detection, units and measurement extraction.
- **Production ML & MLOps:** AWS, S3, EC2, CI/CD, GitLab, Docker, Kubernetes, Helm, model deployment, scalable inference, data pipelines, monitoring-ready architecture.
- **Data & Engineering:** Python, PyTorch, TensorFlow, OpenCV, PySpark, Hive, SQL, PostgreSQL, SQL Server Management Studio, React, React Native, JavaScript.
- **Leadership:** technical lead for enterprise AI delivery, stakeholder translation, end-to-end architecture, production ownership, cross-functional collaboration with engineering/product/client teams.

Education

National University of Singapore - Bachelor of Engineering, Civil Engineering, Honours

Languages

English — Native, Chinese — Native, Dutch — Beginner